UDBHAV SANAGARI

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EDUCATION

SRM Institute of Science and Technology **Expected May 2029**

Bachelor of Technology in Computer Science and Engineering (AI & ML) *Kattankulathur, India*

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming (C++/Java), Discrete Mathematics, Computer Organization

TECHNICAL SKILLS

Languages: Python, C++, Java, Rust, SQL

Frameworks & Libraries: Tauri v2, FastAPI, PyTorch, NumPy, Pandas, Asyncio, ChromaDB, SDL2, FFmpeg

Developer Tools: Git, GitHub, Docker, Linux, Ollama, VS Code, Android SDK, Pytest

Core Competencies: Object-Oriented Design (OOP), Data Structures & Algorithms, RESTful APIs, Local LLM Deployment, System Architecture

PROJECTS

Local Multi-Agent Automation Framework | *Python, FastAPI, SQLite, Ollama, ChromaDB, Asyncio*

- Architected an autonomous, privacy-first multi-agent OS automation engine coordinating an Agent Army (`Manager`, `Worker`, `Vision`) driven by local Ollama models with zero external telemetry.
- Refactored core execution engine with an interactive fast-path self-correction loop, feeding error states and OCR vision feedback back into the worker to dynamically adjust and retry payloads.
- Engineered an asynchronous FastAPI backend with persistent SQLite job tracking (`jobs.db`), in-memory/ChromaDB semantic deduplication for O(1) skill retrieval, and 60+ automated pytest cases.

GCR Simplified — Desktop Grading & Plagiarism Engine | *Rust, Tauri v2, React 19, TypeScript, SQLite*

- Engineered and published a cross-platform desktop application on top of Google Classroom automating assignment evaluation with local-first, FERPA-compliant encrypted credential storage.
- Implemented an offline plagiarism detection engine utilizing the **Winnowing algorithm** and **TF-IDF Cosine similarity**, paired with **SHA-256** binary hashing for instant duplicate file detection.
- Integrated multimodal evaluation via Google Gemini Vision to OCR and grade handwritten student work, auto-generating institutional 4-sheet `.xlsx` gradebooks with class distribution analytics.

LinktoMac — Low-Latency Screen Mirroring & Control | *C++, Java, Python, FFmpeg, VideoToolbox, SDL2*

- Engineered an ultra-low latency (<2–5ms) wireless and USB screen mirroring and remote control system bridging Android 14+ devices to macOS without external cloud dependencies.
- Constructed an end-to-end hardware-accelerated video pipeline utilizing Android `MediaCodec` (HEVC/H.265) streaming over TCP, decoded via Apple `VideoToolbox` with an SDL2/Metal rendering surface.
- Designed a bidirectional input injection protocol for touch gestures and keyboard input via Android `InputManager`, with zero-config mDNS device discovery, 4-digit PIN pairing, and remote SSH tunneling.

CERTIFICATIONS

Claude Code 101 **Anthropic**

- Completed certification on leveraging Anthropic's Claude AI for advanced code generation and problem solving.

EXPERIENCE & EXTRACURRICULARS

Technical Club Member & Contributor **University Technical Club**

- Collaborated in peer coding workshops and team hackathons focusing on core algorithmic problem-solving and rapid prototyping.
- Demonstrated skill in time management and completing software projects in a cooperative team environment.